

A

MAJOR PROJECT REPORT ON

**AI-Driven Real-Time Anomaly Detection and Risk
Mitigation in IoT Network**

Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

BY

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Department of Computer Science and Engineering (Cyber Security)
VIGNANA BHARATHI INSTITUTE OF TECHNOLOGY

(Approved by AICTE, Accredited by NBA, NAAC, Permanently Affiliated to JNTU)

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Academic Year 2024-2025



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(Cyber Security)

CERTIFICATE

This is to certify that the project entitled “**AI-Driven Real-Time Anomaly Detection and Risk Mitigation in IoT Network**” is being submitted by **Ch. Akhil Sampath Vinay (21P61A6209)**, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (Cyber Security)** is a record of bonafide work carried out by them under my guidance and supervision during the academic year 2024-2025. The results embodied in this project report have not been submitted to any other University for the award of any degree or diploma.

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DECLARATION

I, **CH. AKHIL SAMPATH VINAY** bearing hall ticket number **21P61A6209**, hereby declare that the project report entitled **“AI-Driven Real-Time Anomaly Detection and Risk Mitigation in IoT Network”** under the guidance of Mrs. M. Jhansi Rani, Sr. Assistant Professor, Department of Computer Science and Engineering(Cyber Security), **VBIT**, Hyderabad, submitted in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Cyber Security).

This is a record of bonafide work carried out by us and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this project report have not been submitted to any other university or institute for the award of any other degree or diploma.

CH. AKHIL SAMPATH VINAY (21P61A6209)

ACKNOWLEDGEMENT

First and foremost, we wish to express our gratitude towards the institution “**Vignana Bharathi Institute of Technology**” for fulfilling the most cherished goal of our life to do Bachelor of Technology.

It is a great pleasure in expressing a deep sense of gratitude to our Internal guide, **Mrs. M. Jhansi Rani, Sr. Assistant Professor** Department of Computer Science and Engineering (Cyber Security) for her valuable guidance and freedom she gave to us.

We also express our sincere thanks to **Mr. K. Ashok, Associate Professor, Project Coordinator** for his encouragement and support throughout the project.

We are deeply grateful to **Dr. P. Sushma, Head of Department, Department of Computer Science and Engineering (Cyber Security)** for granting us the opportunity to conduct this project.

We take immense pleasure in thanking **Dr. P.V.S. Srinivas, Principal, Vignana Bharathi Institute of Technology, Ghatkesar** for having permitted us to carry out this project work.

Our utmost thanks also go to all the **FACULTY MEMBERS** and **NON - TEACHING STAFF** of the Department of Computer Science and Engineering (Cyber Security) for their support throughout our project work.

CH. AKHIL SAMPATH VINAY (21P61A6209)

ABSTRACT

The rise of web applications has revolutionized digital interactions, but it has also introduced security challenges related to persistent cookies. These cookies, essential for maintaining sessions and improving user experience, can inadvertently expose users to significant risks, including session hijacking, cross-site request forgery (CSRF), and unauthorized data access. This project addresses this gap by introducing the Cookie Analyzer Tool, a comprehensive solution for cookie risk assessment and management.

The Cookie Analyzer Tool systematically captures persistent cookies stored in a browser and evaluates their security using critical attributes such as secure, httponly, samesite, expiry, and domain path. Through a detailed risk analysis, cookies are categorized based on their threat levels, providing users with actionable insights. The tool also facilitates immediate remediation by enabling the deletion of high-risk cookies with a single click and offering security recommendations to mitigate potential threats.

Designed with both technical and non-technical users in mind, the tool features an intuitive interface that enhances usability and accessibility. Users can monitor newly stored cookies in real-time and receive instant notifications of potential risks, empowering them to proactively manage their privacy and security while browsing. By operating entirely on the user's device without internet connectivity, the tool ensures data confidentiality and offers a secure environment for comprehensive cookie management.

In an increasingly web-centric world, the Cookie Analyzer Tool enhances transparency and control over browser cookies, bridging the gap between usability and security. By addressing the hidden risks of persistent cookies, this project contributes to improved user privacy and security, making it an essential tool for modern web browsing.

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Department Computer Science and Engineering (Cyber Security)
COURSE OUTCOMES

Course: Major Project

Class: IV B. Tech II Semester AY: 2024-2025

Course Outcomes

After completing the Projects, the student will be able to:

Code	Course Outcomes	Taxonomy
C424.1	Identify and state the problem precisely to prepare the abstract	Understand
C424.2	Analyze the existing system, and outlining the proposed methodology for effective solution	Analyze
C424.3	Use various modern tools for designing applications based on specified requirements	Apply
C424.4	Develop applications with adequate features and evaluate the application to ensure the quality	Create
C424.5	Prepare the document of the project as per the guidelines	Create

PROGRAM OUTCOMES

PO 1: Engineering Knowledge:

Apply the knowledge of mathematics, computer science, engineering fundamentals, and cybersecurity principles to develop solutions for complex problems in chat monitoring and threat detection.

PO 2: Problem Analysis:

Identify, formulate, research literature, and analyze complex issues related to online communication threats using first principles of data science, natural language processing, and cybersecurity.

PO 3: Design/Development of Solutions:

Design and implement solutions for chat monitoring and threat detection systems that meet user needs while addressing public safety, privacy, and ethical considerations.

PO 4: Conduct Investigations of Complex Problems:

Use research-based knowledge to investigate issues such as phishing, spam, and abusive communication in chat systems, employing data analysis and machine learning techniques to provide valid conclusions.

PO 5: Modern Tool Usage:

Create, select, and apply modern IT tools, such as machine learning algorithms, natural language processing frameworks, and real-time analytics, for effective threat detection in chat systems, while understanding their limitations.

PO 6: The Engineer and Society:

Assess societal, legal, and cultural impacts of chat monitoring systems and ensure compliance with regulations and ethical practices in their implementation.

PO 7: Environment and Sustainability:

Understand the environmental impact of deploying scalable chat monitoring systems and advocate for sustainable practices in system development and operation.

PO 8: Ethics:

Apply ethical principles and commit to professional responsibilities by ensuring user privacy and transparent operation of chat monitoring and threat detection systems.

PO 9: Individual and Team Work:

Work effectively as an individual or as part of a multidisciplinary team to design, develop and deploy chat monitoring solutions that address cyber security challenges.

PO 10: Communication:

Communicate effectively with stakeholders, including presenting technical reports, designing user-friendly dashboards, and creating comprehensive documentation for chat monitoring systems.

PO 11: Project Management and Finance:

Demonstrate project management skills by planning, developing, and deploying chat monitoring systems within budgetary and resource constraints, both independently and as a team leader.

PO 12: Life-long Learning:

Recognize the importance of continuous learning to stay updated with advancements in natural language processing, machine learning, and cyber security, ensuring the relevance of chat monitoring and threat detection systems in a rapidly evolving technological landscape.

PROGRAM SPECIFIC OUTCOMES

PSO1:

Demonstrate expertise in chat monitoring techniques, including proficiency in detecting vulnerabilities, identifying malicious activities, and mitigating potential threats within communication platforms.

PSO2:

Showcase mastery in collecting, preserving, and analyzing chat data to support the identification of threats, legal investigations, and resolution of cybercrime cases.

PSO3:

Exhibit managerial skills and knowledge, combined with proficiency in deploying cutting-edge technologies, to address future challenges in ensuring the security of digital communication systems for individuals, organizations, and society.

VIGNANA BHARATHI INSTITUTE OF TECHNOLOGY

Department Computer Science and Engineering (Cyber Security)

COs Mapping with PO/PSO

Project Title : AI-Driven Real-Time Anomaly Detection and Risk Mitigation in IoT Network

Name of the Supervisor : Mrs. M. Jansi Rani

Batch Details :

S. NO	Regd.No.	Student Name	Technology
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2	21P61A6216	G. TRISHA	CYBER SECURITY
3	21P61A6255	S. PRAVEEN	CYBER SECURITY

Note: Write your domain name in technology field (ex. ML, IOT, BC, Security, Cloud etc.)

CO-PO Mapping for Major Project:

High -3 Medium -2 Low-1

PO/CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
C424.1	3	3	2.5	3	3	2.5	2	2.5	3	3	3	3	3	3	3
C424.2	3	3	3	3	2.5	3	2.5	3	3	3	3	3	3	3	3
C424.3	3	2.5	3	2.5	3	3	3	3	3	3	2.5	3	2.5	3	2
C424.4	2.5	3	2.5	2	3	3	3	2.5	3	3	2.5	2	2	2	3
C424.5	2.5	2.5	2	2.5	3	2	2	3	2	2	3	2.5	2.5	2	2
Avg	2.8	2.8	2.6	2.6	2.9	2.7	2.5	2.8	2.8	2.8	2.8	2.7	2.6	2.6	2.6

Guide Signature

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